BrainHealth

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What Makes a Brain Healthy?

“Brain health is the state of brain functioning across cognitive, sensory, social-emotional, behavioural and motor domains” -World Health Organization

With different physical health statuses, environments, economic situations, brain and nervous system conditions, brain growths or damage to brain structure, are we able to have a quantitative score for qualitative factors?

If we did, we could take advantage of the great leaps in compute and data analytics that have been made to better understand our brain health

BHI:

The BrainHealth Index (BHI) is a holistic measure of brain health developed by the Center for BrainHealth at UT Dallas. It is a composite score built from 22 behavioral assessments using a proprietary machine-learning algorithm

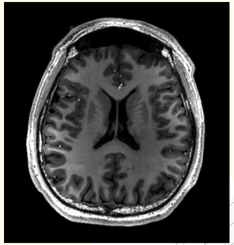
It measures three core domains:

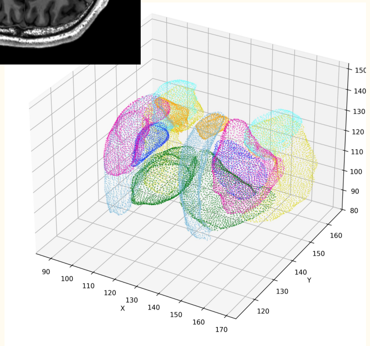
- (F1) Clarity: Strategic attention, reasoning, memory, and innovation. How effectively the brain processes and applies information
- (F2) Emotional Balance: Depression, anxiety, and stress levels. Emotional steadiness under difficult conditions
- (F3) Connectedness: Social support, happiness, resilience, and sense of purpose

The index can track how an individual's brain health changes and improves over time

Can Brain Anatomy Explain BrainHealth?

- BrainHealth scores change over time
- MRI captures brain structure
- Can anatomy help explain BHI variation?





Features → BHI?

3D Mesh Analysis

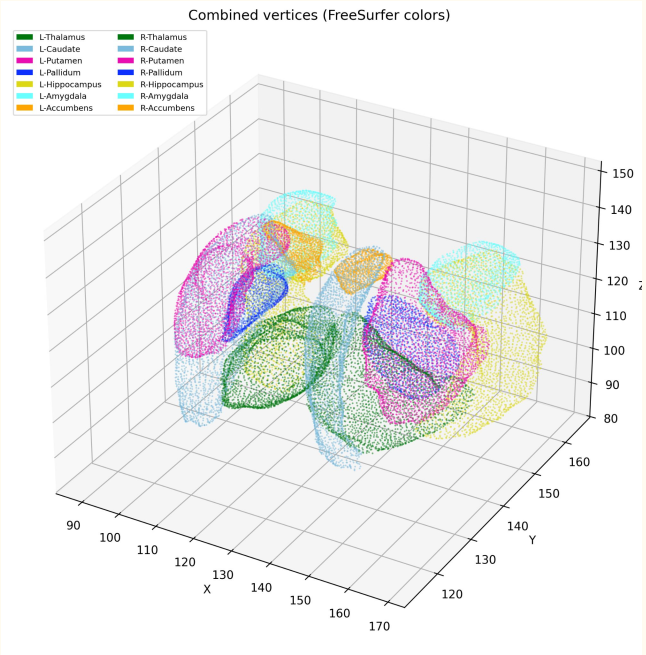
Dataset

69 unique participants

240 Scans

Brain Health Index (BHI) and factor scores:

- F1: Social Interaction & Mental Fortitude
- F2: Mood & Well-Being
- F3: Cognition & Empathy



Meshes (Vertices and Faces) of 14 Different Subcortical Structures

Previous Work in Volume Based Analysis

Results:

- Weak negative correlation between brain volumes and BHI scores (larger brain volumes associated with higher BHI)
- None of the correlations rose to the degree of significance
    - BHI and F3 were the closest, but still not enough

Shape Descriptor Computation

For every structure, six descriptors were computed:

- Volume (Size of structure)
- Surface Area (Total mesh surface)
- Sphericity (Similarity to a sphere)
- Compactness (Volume-to-surface efficiency)
- Mean Curvature (Surface irregularity)
- Convex Hull Ratio (Degree of convexity)

Idea: Shape descriptors may capture anatomical changes not reflected by volume alone

Results: Shape Descriptors

- Shape descriptors outperformed volume alone in associating with BrainHealth Index (BHI)
- Sphericity and compactness showed significant relationships with:
  - BHI ($p \approx 0.04$)
  - F2: Mood & Well-being ($p \approx 0.05$)
  - F3: Cognition & Empathy ($p \approx 0.01$)
- Volume-based analysis was only significant for F3
- LOSO cross-validation R^2 remained near zero (—0.086 to +0.032)

Not really statistically significant but there is more correlation than volume.

Vertex-Based Morphometry of Subcortical Brain Structures

Key Question: Can localized shape changes in subcortical brain structures be better indicators of brain health instead of traditional volume measurements?

Our Approach: Vertex-Based Morphometry examines thousands of individual surface points on brain structures, allowing precise mapping of localized anatomical changes

Hypothesis: Specific subcortical regions would show distinct morphological relationships with both the latent features and overall brain health indices.

Methods

Morphometric Features:

1. Local Surface Area: Measures the amount of surface surrounding each vertex which reflects localized expansion or atrophy
2. Mean Curvature: Quantifies how much the surface bends or folds
3. Radial Distance from Centroid: Distance from the structure's center to each vertex, acting as a local indicator of structural size or volume change

Outcomes Examined:

Brain Health Index

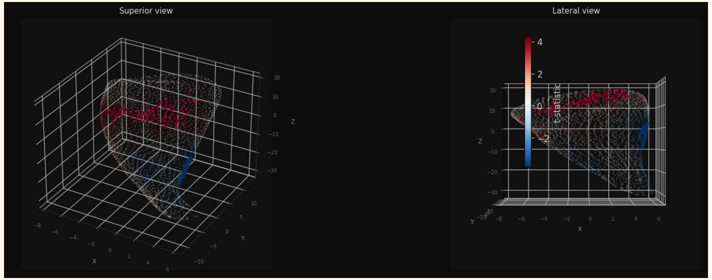
Latent Factors:

- F1: Social interaction & mental fortitude
- F2: Mood & well-being
- F3: Cognition & empathy

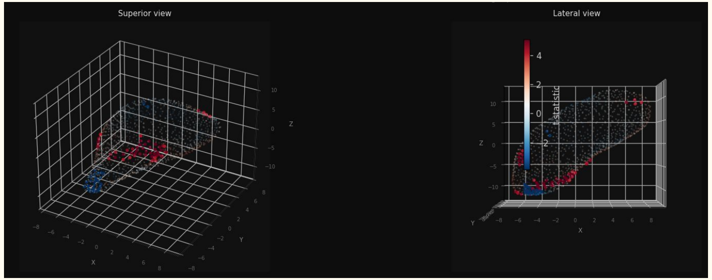
Results - Associations with Overall BHI:

Six structures showed significant surface area reductions associated with lower BHI: left amygdala, right putamen, left hippocampus, left caudate, right pallidum, and left pallidum

3 structures showed significant mean curvature associations with BHI: left amygdala, right pallidum, and right hippocampus but at lower percentages (0.04%-0.8% compared to previous (2.2%-13.8%))



Area Association Left Caudate

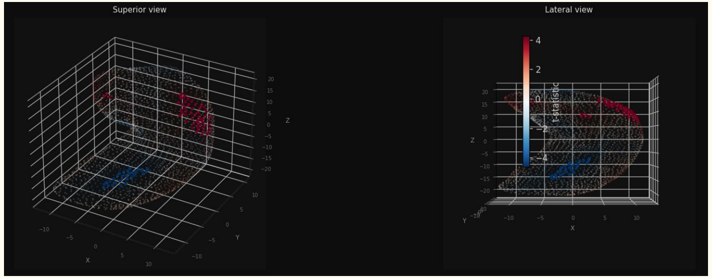


Radial Association Right Pallidum

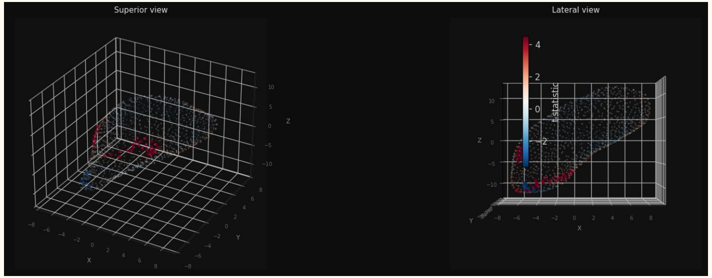
Results - Associations with F1:

The right putamen, left hippocampus, right amygdala, and left caudate all showed localised surface area reductions with lower F1.

For radial distance, the right pallidum showed a positive association, while the right hippocampus, left accumbens, and right putamen showed negative associations.



Area Association Left Hippocampus



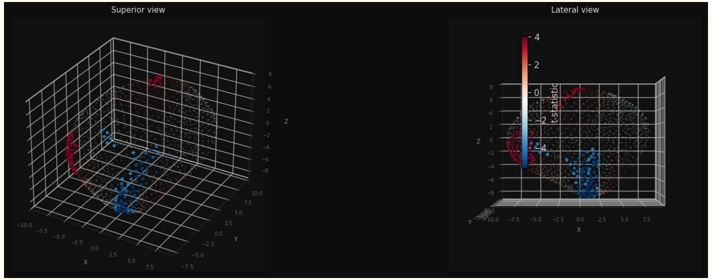
Radial Association Right Pallidum

Results - Associations with F2:

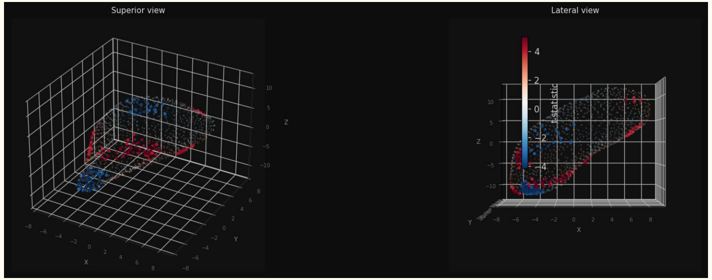
F2 yielded the most spatially extensive associations in the amygdalae and pallida, with the left amygdala being the strongest single finding in the dataset

For radial distance, the right pallidum showed a positive association while the left amygdala showed a negative association.

Mean curvature effects were observed in the both amygdalae, left pallidum, left putamen, and right hippocampus, though to a lesser extent



Area Association Left Amygdala



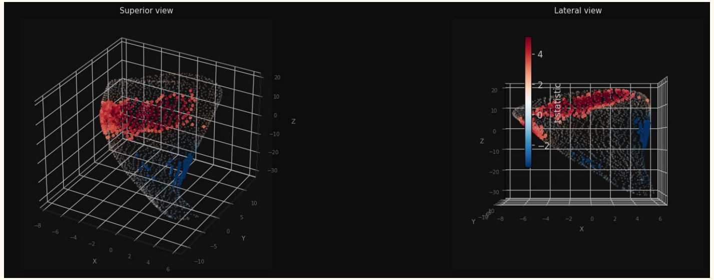
Radial Association Right pallidum

Results - Associations with F3:

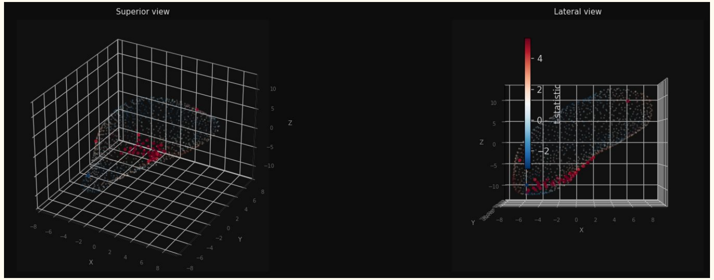
The most spatially extensive finding was a positive surface area association in the left caudate

For radial distance, the right pallidum showed the strongest association while the right caudate showed a negative association

Mean curvature effects were generally contained to isolated vertices in both amygdalae and pallida, as well as the right thalamus and left hippocampus.



Area Association Left Caudate



Radial Association Right Pallidum

MRI Pipeline

Basic Overview

- We apply preprocessing to 499 MRI images and feed these images into the BrainIAC model
- BrainIAC generates features in the latent space, and outputs these results
- We ran Linear Regression, PCA + Linear Regression, Ridge Regression, and CCA on these features to determine if there was any correlation/relationship between them and BHI or any of its components

What is BrainIAC?

- BrainIAC (Brain Imaging Adaptive Core) is a foundation model designed for generalized analysis of human brain MRI data
- It is a ViT (Vision Transformer) model that uses self-supervised contrastive learning to learn how to construct features from MRI scans
- Pretrained on 32,015 multiparametric brain MRIs and validated across a total of 48,965 scans
- Developed for use in scenarios where only a small number of labeled data exist

Data Preprocessing

Raw T1-weighted MRI scans are preprocessed using the BrainIAC pipeline before feature extraction:

- Image Registration — N4 bias field correction and rigid registration to a common atlas template using Mattes Mutual Information
- Skull Stripping — non-brain tissue removed using HD-BET (deep learning-based brain extraction)
- Normalization — resized to $96 \times 96 \times 96$ voxels, intensity normalized channel-wise across nonzero voxels

The preprocessed scans are then passed through the BrainIAC ViT-B backbone in inference mode to produce one 768-dimensional feature vector per scan

The Feature Dataset

Each scan's 768-dim vector is paired with that visit's health scores and demographics to form one row in our dataset

SubjectID	Timepoint	Feature_0	Feature_1	...	BHI	F1	F2	F3	Age	Sex
PIL007	T1	1.701	0.350	...	694	481	513	488	24	M
PIL007	T2	1.724	0.404	...	728	514	491	580	24	M
PIL007	T3	1.748	0.397	...	707	505	457	559	24	M

- 108 subjects, up to 7 timepoints → 499 total scans
- 768 features per scan from BrainIAC
- 378 consecutive visit-pairs for longitudinal analysis
- 13 observations with missing health scores excluded

The 768 features as the representation of brain anatomy, correlated and regressed against BHI, F1, F2, and F3. Age and Sex are included as covariates — used to control for demographic effects and verify that any signal found is genuinely anatomy-related

We analyze F1 (social connectedness), F2 (mood), and F3 (cognition) separately rather than relying solely on the composite BHI score as each captures a distinct domain of brain health and may manifest differently in brain structure.

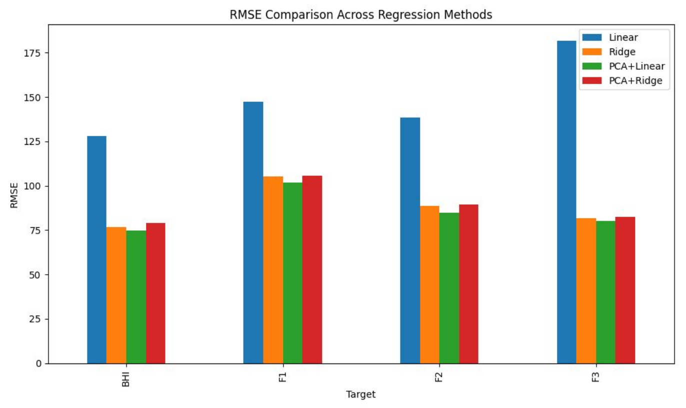
What Is PCA?

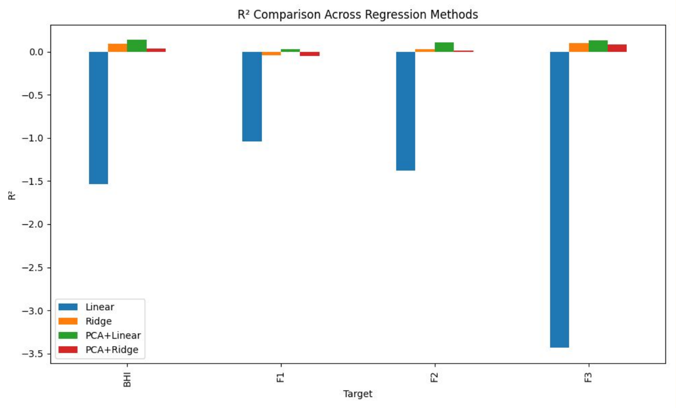
- Finds new axes (principal components) that capture maximum variance in the 768-dimensional feature space
- Components are ordered: PC1 explains the most variance, PC2 the next, and so on
- Compresses features into a smaller set of uncorrelated variables before fitting regression
- Directly addresses the issue of numerous features (700+ generated by BrainIAC) by keeping only the directions where data actually varies
- Here, components explaining 95% of variance were retained

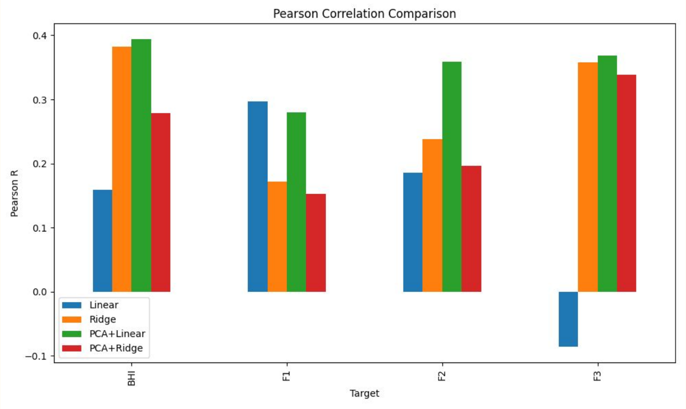
What is CCA?

- Finds paired linear combinations (one from brain features (X), one from health targets (Y)) that are maximally correlated with each other
- Unlike per-target regression, CCA treats BHI, F1, F2, and F3 jointly
- Each canonical dimension captures a different mode of shared covariation between the two spaces
- A high canonical correlation on dimension 1 means the strongest shared structure between brain anatomy and health outcomes is well captured
- Dimensions with near-zero or negative correlations (dims 3 & 4 here) indicate those modes carry no real shared signal

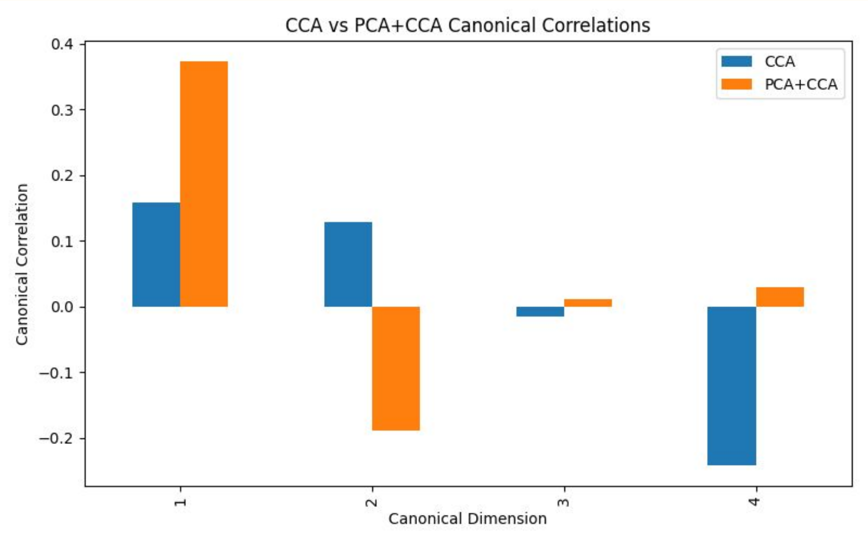
Results


RMSE Comparison Across Regression Methods
RMSE
Target
BHI
F1
F2
F3
Linear
Ridge
PCA+Linear
PCA+Ridge
0
25
50
75
100
125
150
175


R² Comparison Across Regression Methods
R²
Target
BHI
F1
F2
F3
Linear
Ridge
PCA+Linear
PCA+Ridge
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5


Pearson Correlation Comparison
Pearson R
Target
BHI
F1
F2
F3
Linear
Ridge
PCA+Linear
PCA+Ridge
0.4
0.3
0.2
0.1
0.0
−0.1

Results cont.



Conclusions

- Linear regression
  - Completely fails across all targets, R^2 as low as -3.4 for F3, RMSE up to 181.8
  - Massively overfits with 768 raw features on 108 subjects
  - Produces a negative Pearson R for F3, meaning predictions are worse than chance
- Ridge regression
  - Recovers to small positive R^2 for BHI and F3 (0.09-0.10); still negative for F1
  - RMSE drops dramatically vs. linear (about 76-105), showing regularization helps substantially
  - Pearson R is strongest for BHI (0.38) and F3 (0.36), F1 remains the weakest target
- PCA + linear regression
  - Best overall performer, highest R^2 and Pearson R across nearly every target
  - RMSE is marginally lower than Ridge across all four targets
  - Compressing to 95% variance components before fitting cleanly solves the dimensionality problem
- PCA + ridge regression
  - Underperforms PCA + linear across all metrics, adding ridge on top of PCA offers no benefit
  - PCA already removes collinearity, so ridge's regularization has little left to contribute
  - R^2 and Pearson R are notably lower than PCA + linear, especially for BHI and F1
  - RMSE is comparable to plain Ridge, suggesting the PCA compression did not help here either

Conclusions cont.

- CCA
  - Finds only weak shared structure between MRI features and behavioral outcomes, with the strongest canonical correlation around -0.24
  - Later dimensions become unstable, including a moderate negative correlation (-0.24) in the fourth canonical variate pair
  - High-dimensional feature space (768 features, 108 subjects) likely limits CCA's ability to estimate robust canonical directions
- PCA + CCA
  - Produces the strongest overall multivariate association, increasing the first canonical correlation from 0.16 to 0.37
  - PCA removes redundant and noisy feature variation, allowing CCA to focus on the dominant shared signal
  - Higher-order canonical dimensions contribute little additional information, suggesting most meaningful covariance is concentrated in the first component
  - Eliminates the unstable negative correlations observed in raw CCA, yielding a more interpretable and robust canonical solution

Method 1 — Pooled (Cross-Sectional) Correlation

$$r = \frac{\Sigma\,(\text{Feature\_i} - \text{mean})(\text{BHI} - \text{mean})}{\sqrt{[\,\Sigma(\text{Feature\_i} - \text{mean})^2 \cdot \Sigma(\text{BHI} - \text{mean})^2\,]}}$$

- Computes Pearson r between each of the 768 features and the health score across all 499 rows
- $r = +1$ means the feature and health score move perfectly together; $r = 0$ means no relationship
- Fast and simple but treats every scan independently, does not account for the fact that scans come from the same people over time
- Dominated by differences between people (age, genetics, baseline anatomy) rather than changes within one person

Method 2 — Within-Subject Δ Correlation

$$\Delta\text{Feature} = \text{Feature}(T+1) - \text{Feature}(T)$$
$$\Delta\text{BHI} = \text{BHI}(T+1) - \text{BHI}(T)$$

$$r = \text{corr}(\Delta\text{Feature}, \Delta\text{BHI}) \quad \text{across 378 visit-pairs}$$

- For each subject, computes the change between consecutive visits then correlates those changes
- Removes everything stable about a person — age, sex, and baseline anatomy all cancel out
- What remains is purely the longitudinal signal: when a person's brain representation shifts, does their health score shift with it?
- This is the scientifically correct method for a digital twin, which simulates change within an individual

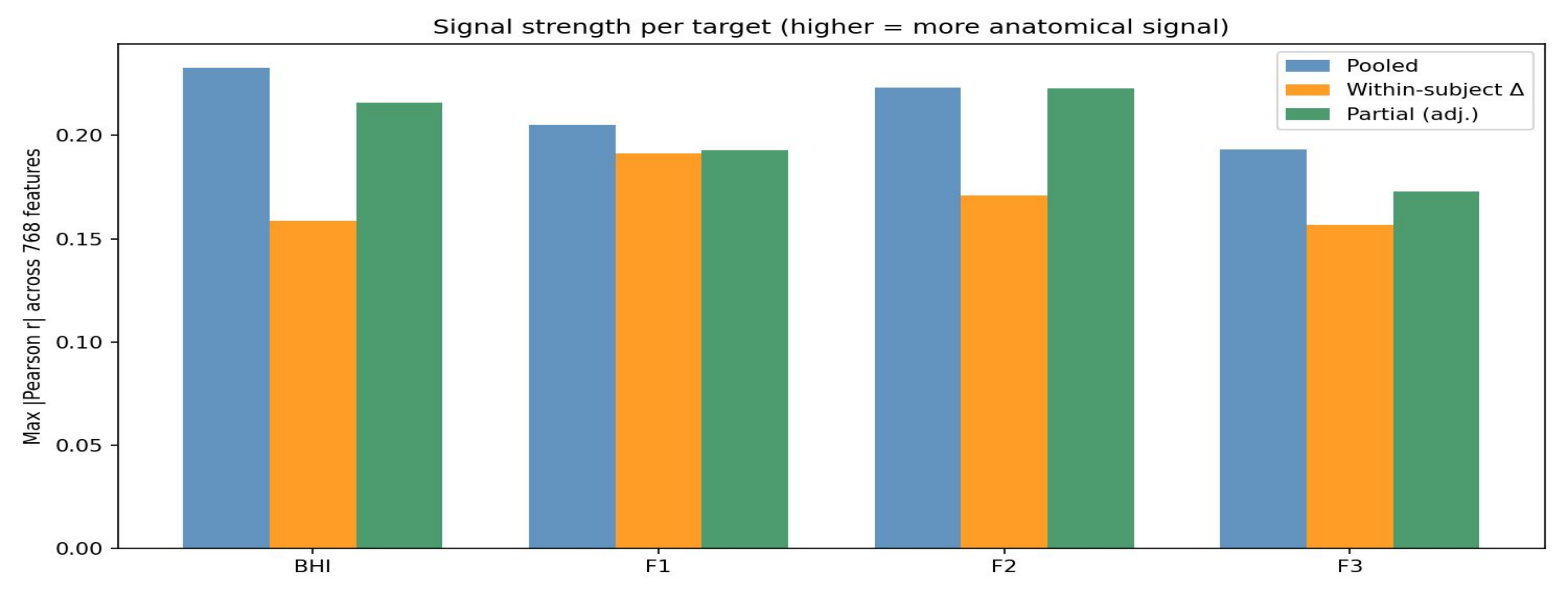
Method 3 — Partial Correlation (Age + Sex Adjusted)

$$C = [1, Age, Sex]$$
$$X\_resid = X - C(C^T C)^{-1} C^T X$$
$$y\_resid = y - C(C^T C)^{-1} C^T y$$

$$r\_partial = corr(X\_resid, y\_resid)$$

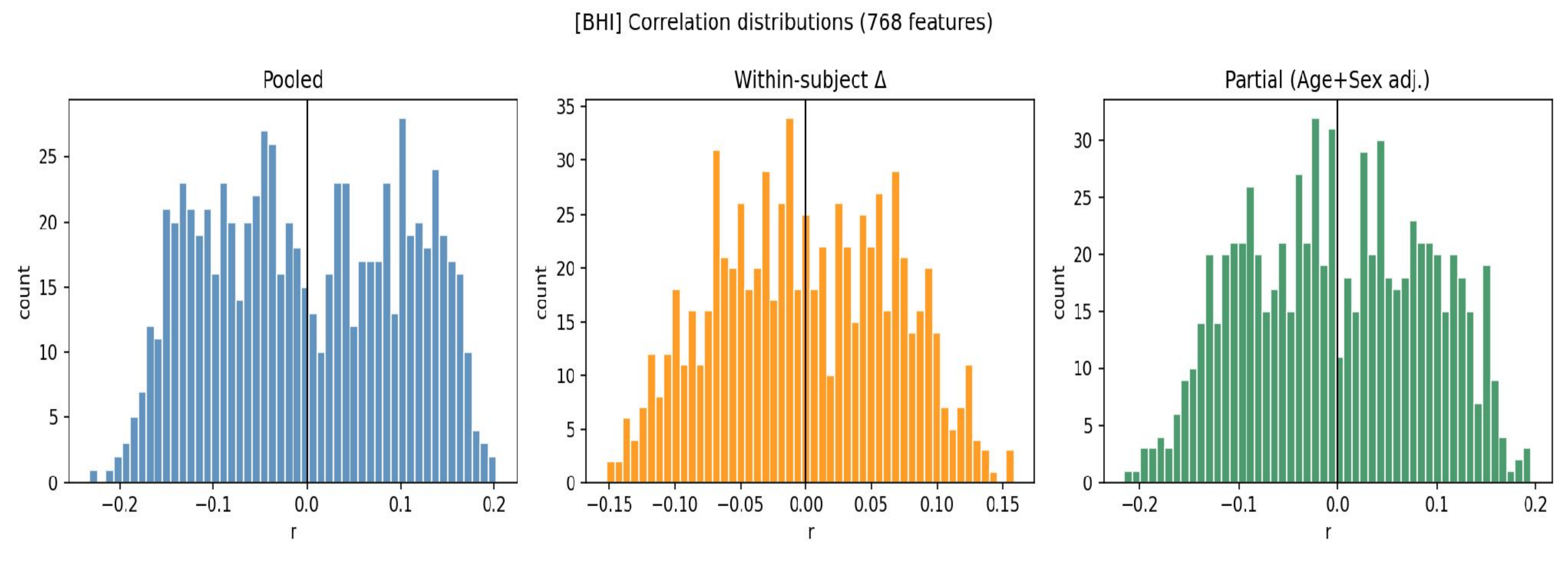
- Regresses Age and Sex out of both the features and the health score, then correlates the residuals
- Tests whether the pooled signal is simply an artifact of older people having both lower BHI and different anatomy
- Result close to method 1, Age and Sex explain almost none of the correlation
- Confirms health signal is not driven by demographics

Results — Signal Strength Across All Targets



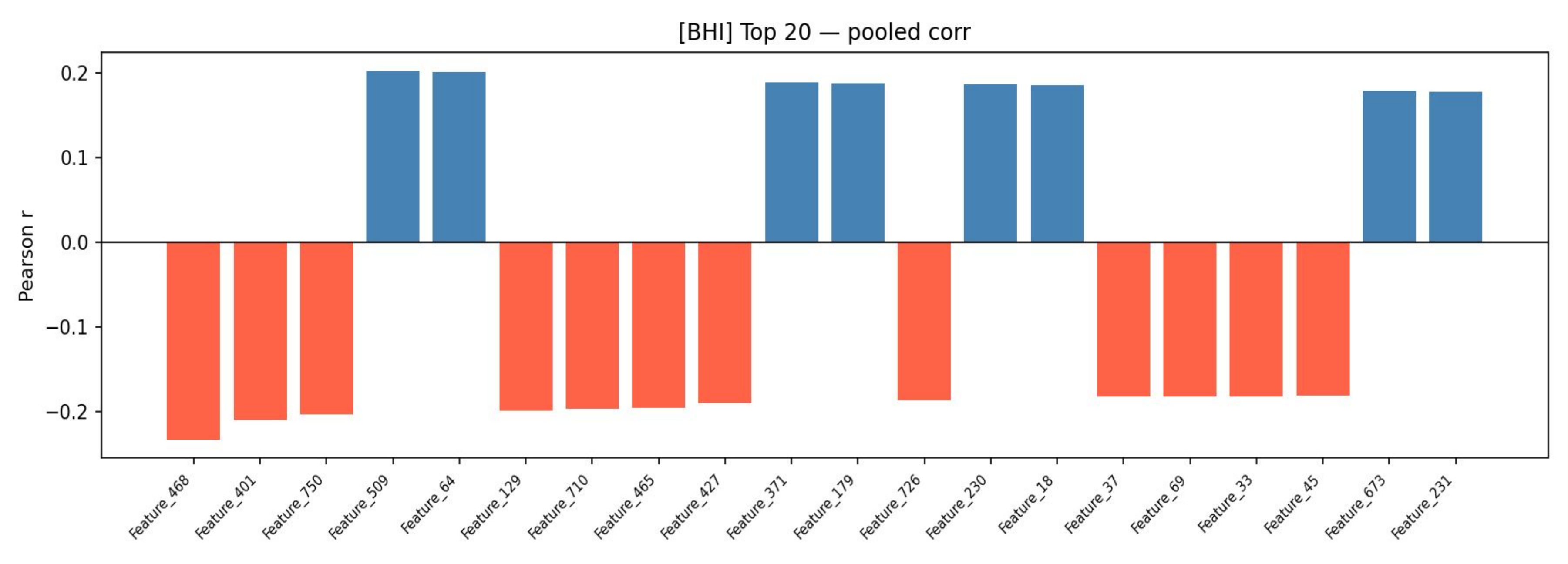
- BHI has the strongest cross-sectional signal (pooled r = 0.233) — expected as the overall composite
- F1 (social) has the strongest within-subject delta correlation (r = 0.191) — best longitudinal tracker
- F3 (cognitive) is consistently the weakest target across all three methods
- No single feature has a strong correlation — the signal is weak and distributed across all 768 dims

Results — Correlation Distributions (All 768 features)



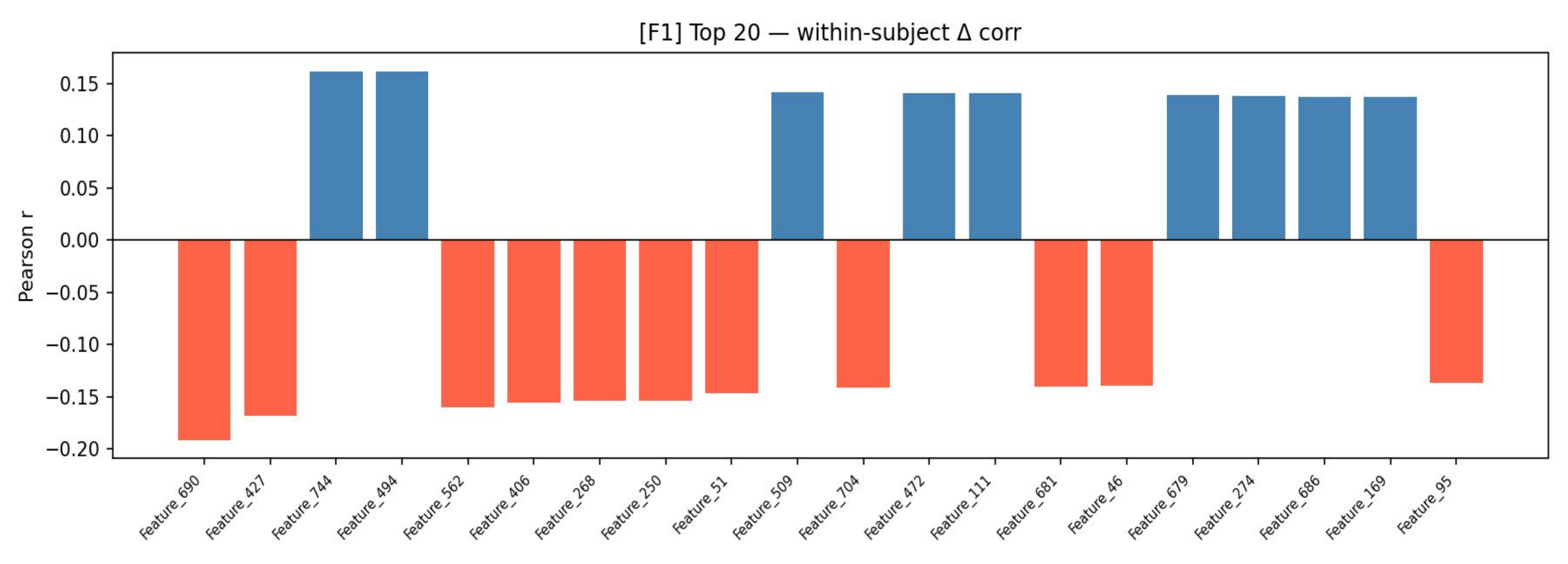
- No single feature dominates, correlations spread broadly around zero across all 768 features
- Within-subject Δ (orange) is narrower and tighter
- Partial (green) is not very different from pooled (blue)

Results — Top 20 Features, BHI Pooled Correlation



- Shows which features differ between people who have high vs low BHI overall
- Blue = positive correlation (higher feature → higher BHI); Red = negative correlation

Results — Top 20 Features, F1 Within-Subject



- Shows which features actually shift when a person's F1 score changes between visits
- F1 (social) has the highest max within-subject r = 0.191 of all four targets (BHI, F1, F2, F3)
- Top features here are largely different from the BHI pooled list — social health has its own distinct anatomical signature in latent space

Conclusions

- The BrainIAC latent features contain real but weak anatomical signal for brain health (max $|r| \approx 0.23$) — confirmed genuine by partial correlation
- The signal is distributed across all 768 dimensions — no single feature dominates, meaning the health axis in latent space is a collective direction
- Analyzing F1, F2, F3 separately is more informative than BHI alone — each factor has a distinct anatomical footprint that gets diluted in the composite score
- F1 (social) is the strongest longitudinal target — highest within-subject delta correlation (0.191) and strong direction alignment (0.793)